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ENERGY COOP · EMBASSY

Energy communities

CleantechHUB as a partner against energy poverty: public data, O&M cooperatives, and Colombia's cleantech ecosystem.

Albert Kobus · 11 Nov 2026 · cleantechhub.net

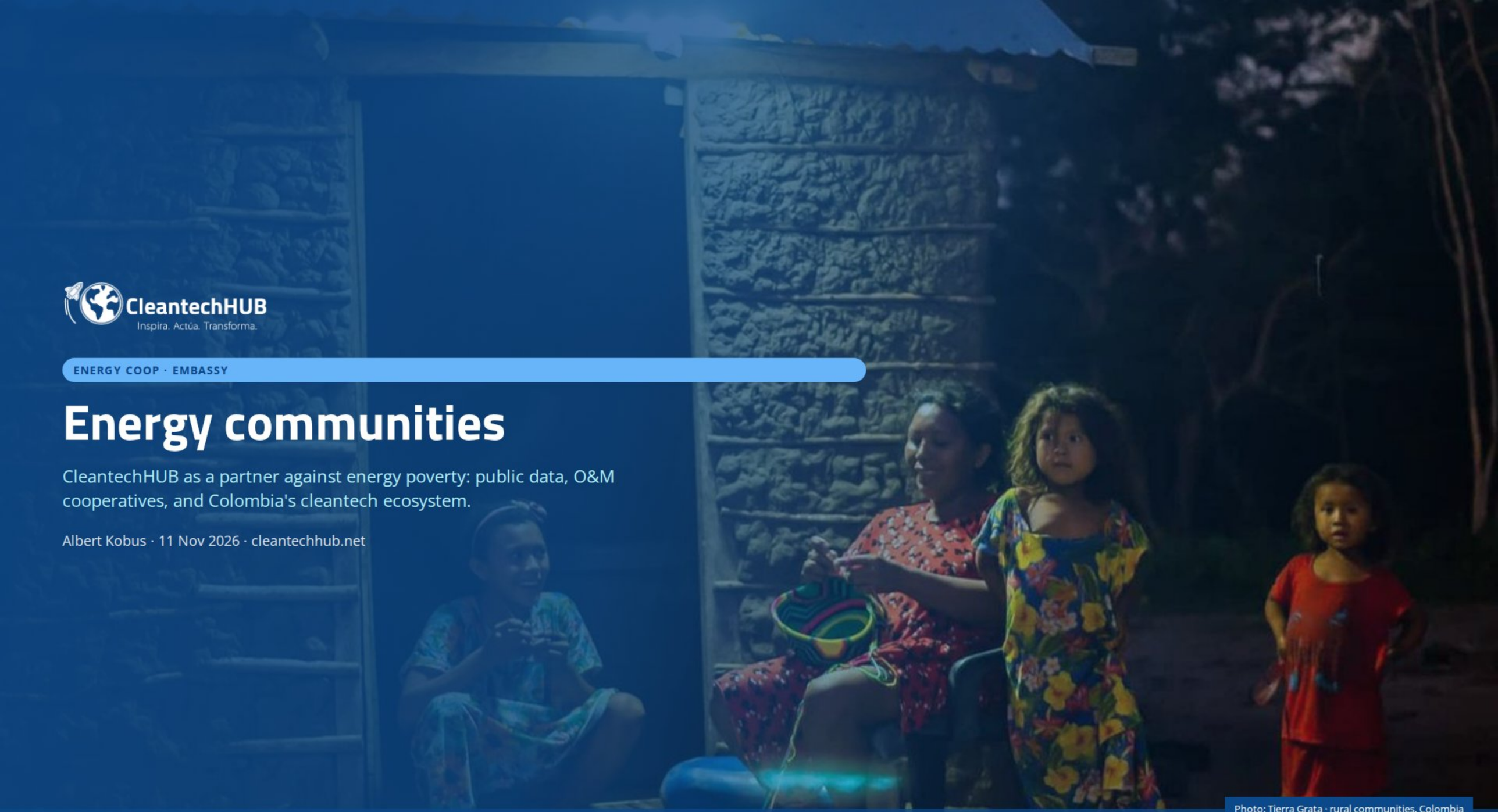
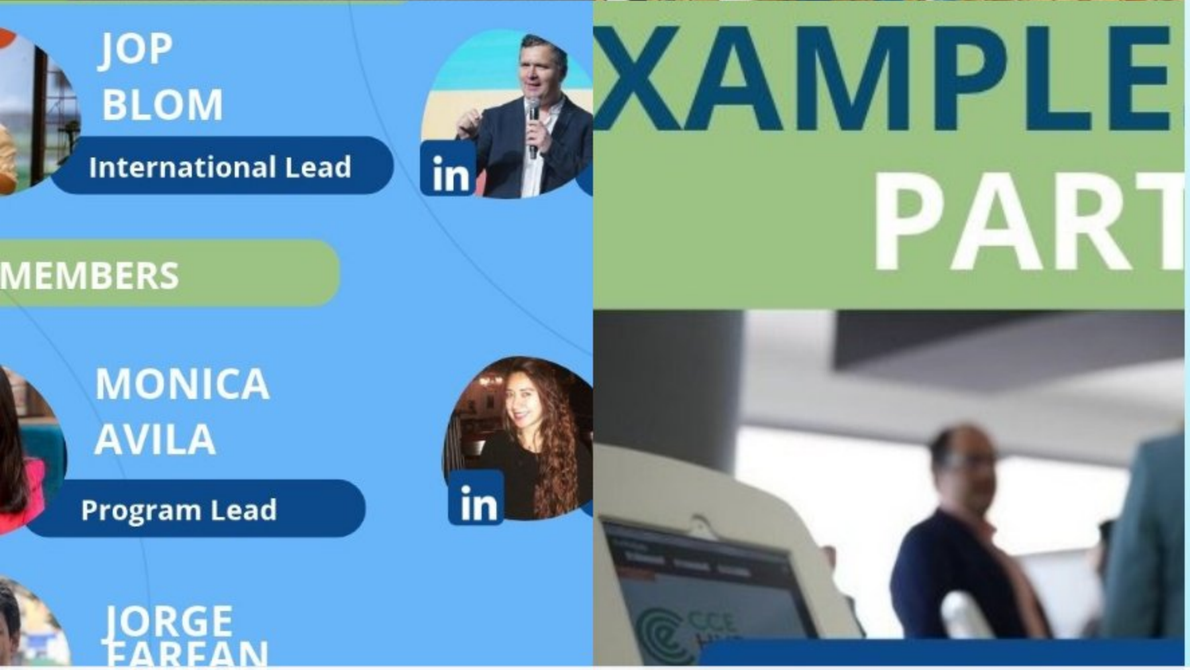


Photo: Tierra Grata · rural communities, Colombia





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CURRENT

CleantechHUB

Climate innovation hub in LATAM: cleantech startups, corporates, investors, and government.

2019
Founded (cleantechhub.net)

5
Countries cited: CO, PE, SV, GT, CR

3
Platforms: web · wiki · Nexus

LATAM
Regional focus (schema.org)

Source: cleantechhub.net JSON-LD Organization, Oct 2026.









chHUB
Transforma.



r 2025
bastián



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CURRENT

ClimateLaunchpad CLP26

National Organizer for the world's largest green business ideas competition — pipeline to Nexus.

5
2026 campaign countries: CO, PE, SV, GT, CR

CLP
National Organizer · CleantechHUB

6–8
Week call window (clp26-brand)

decks.cleantechhub.net/cth-clp26

ClimateLaunchpad global

Source: skills/clp26-brand/SKILL.md · CLAUDE.md.



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Nexus

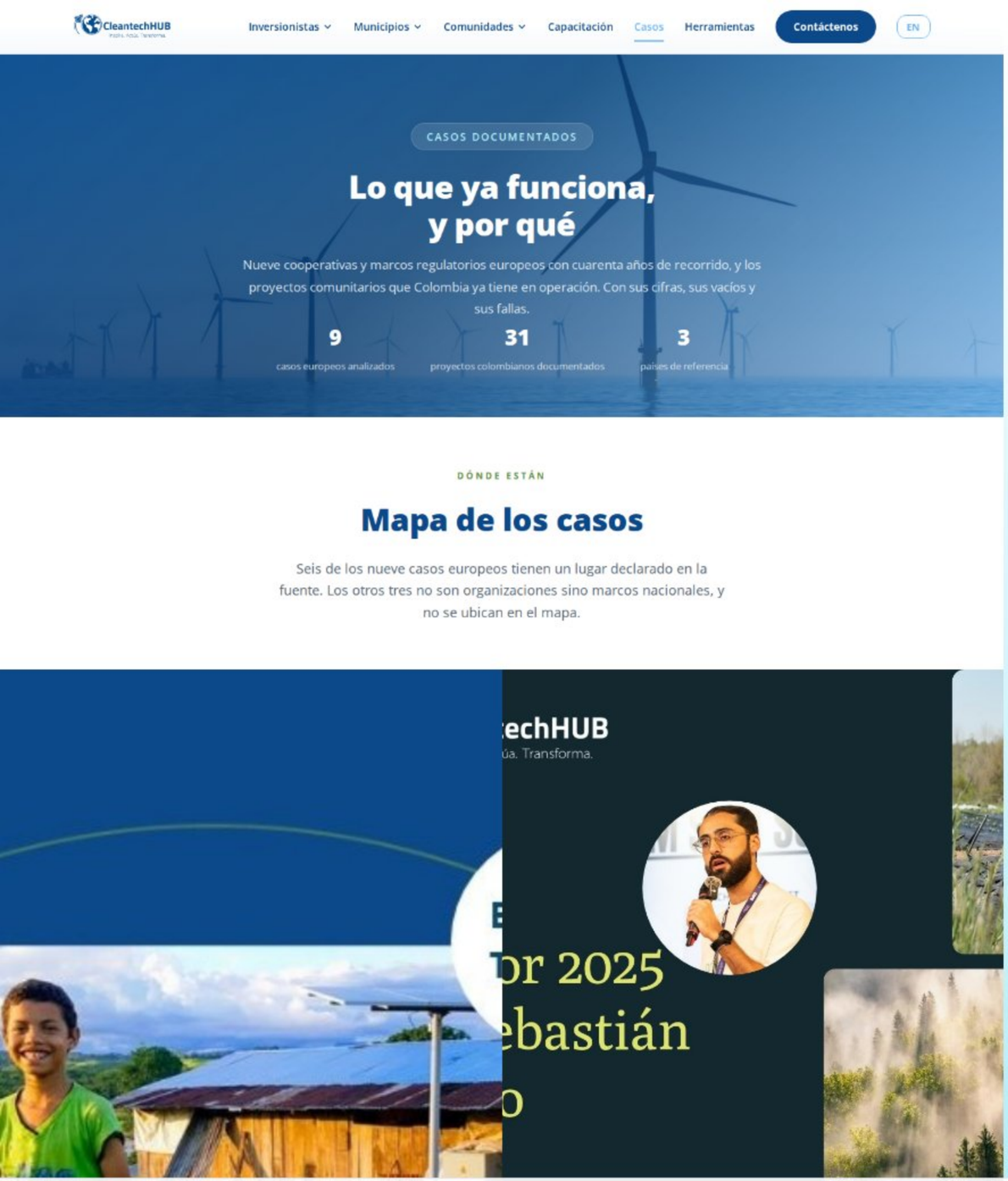
Bilingual cleantech startup portfolio — profiles at nexus.cleantechhub.net.

5
Energy Coop startups (brochure-v2)

EN·ES
Bilingual startup profiles

Sunco · SEOS · Soluna · Tierra Grata · TuWatt

Source: skills/nexus-onepager · brochure-v2.



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CURRENT

Energy Coop

Energy cooperatives: social layer and O&M where CAPEX and an implementer exist — no MW funds.

15.4%
National IMPE 2024 (~8.1M people)

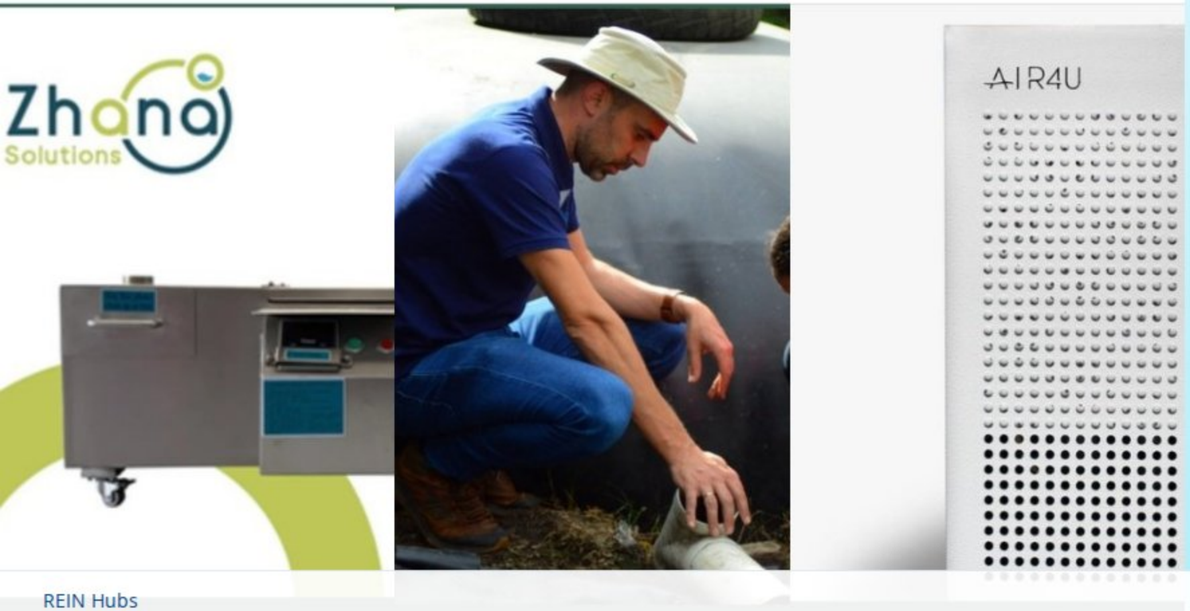
2.9M
People to reach 2030 target

898
kWp Sunco · DISPAC La Guajira[9]

1,947
Tierra Grata energy solutions[11]

Source: brochure-v2 · IMPE 2025 · energiacoop.cleantechhub.net/casos/





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REIN Hubs

LATAM network of physical hubs — cleantech activation, incubation, and acceleration with PVBLIC Foundation.

5+
Active LATAM hubs (REIN deck)

CLP
Global ClimateLaunchpad pathway

CTH
Delivery partner · no $ figures here

PVBLIC Foundation | Kintec Academy

Source: decks/rein-academic-city-ghana — AGENTS.md (no REIN dollar amounts).



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PRIOR

P4G / MubOn

Competitive P4G proposals and MubOn packs in CTH archive — strict co-finance rules, no invented figures.

50/50
Typical P4G co-financing (cth-grant)

Grant
UNDP graph · P4G/MubOn (Drive)

P4G · MubOn · UNDP context

Source: skills/cth-grant/SKILL.md · docs/grant-graph/spec.md.

ENERGY COOP

The gap and OPEX

15.4%
IMPE 2024 · ~8.1M

9.5%
2030 target · ~2.9M

75.9%
Rural ICEE 2024

63.1%
IMPE La Guajira

When do we close the 2.9M-person gap?

Observed pace ~300k/year

~2034

IMPE target ~490k/year × 6 years

2030

Contraloría
Microgrid O&M not remunerated.

Amazonas
~$30,000M royalties in idle PV.

CREG
Abandoned rural PV without management.

CTH: *bankable* social layer and OPEX with committed CAPEX and implementer.



PROPOSAL

Cooperative + ecosystem

Community O&M · cooperative governance · IMPE dimension KPIs (does not replace ICEE).

SN	SE	SL	TG	TW
Sunco	SEOS	Soluna	Tierra Grata	TuWatt
898 kWp[9]	+75 SOLvers[10]	ZNI	1,947[11]	Coops

Candidate: Corozal, Sucre (IMPE 43.1%) — not a PASO energy project.

Data lab (illustrative)



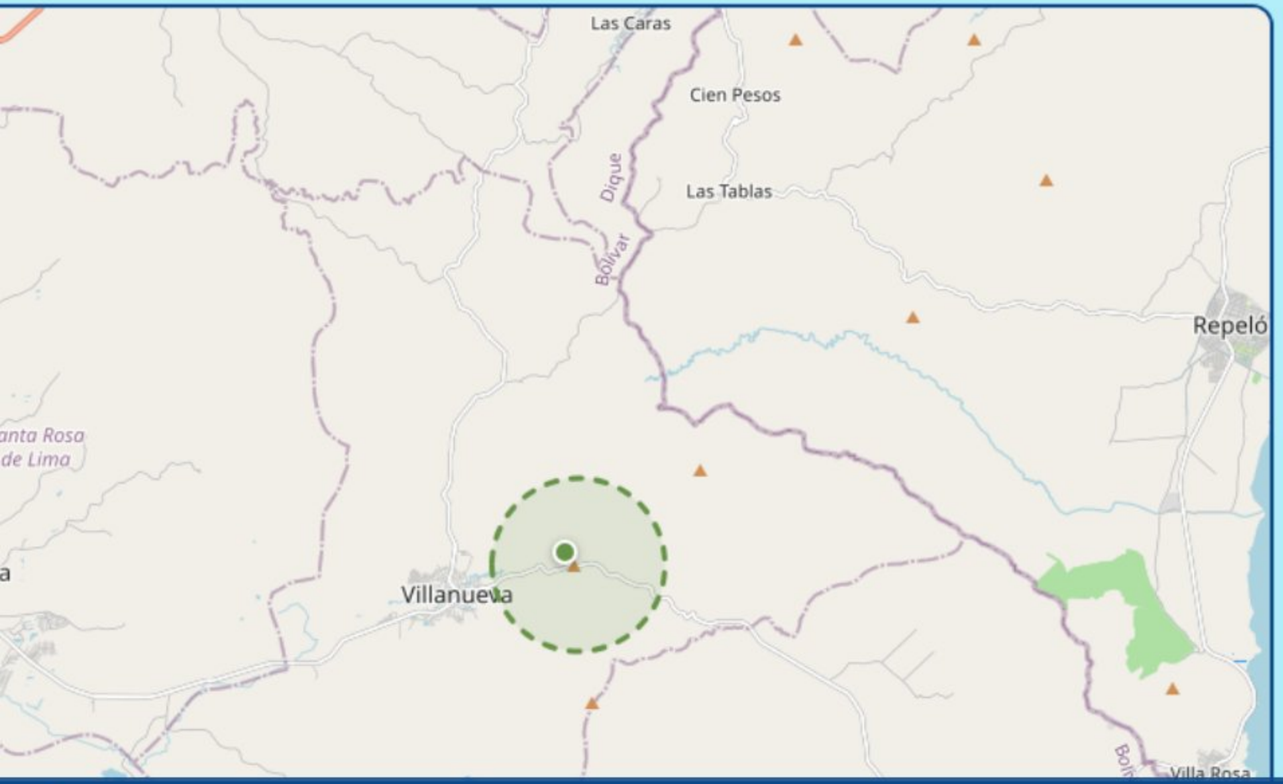
~1,000	100	20	10
Public cross-cut	Scoring	Validation	Co-op wave

: from committee to cooperative

ty solar energy in vereda Paiva, Bolívar (~450 inhabitants), IE Chiricoco. NOT FENOGE Map location **approximate**.

l energy poverty (IMPE, index on persons): 15.4% · ~8.1 M people (2024)
MPE 2025 (PDF) · Nota IMPE, nov 2025

ectricity coverage 75.92% vs national 93.12% (UPME ICEE 2024)
EE 2024 (UPME) · UPME, noticia 2024



PUBLIC SOURCE

Community solar project with local committee and IE Chiricoco (linked public sources).

Noticia Paiva (2025) · Guía del Gas

CTH DESIGN PROPOSAL

Guided transition: community committee → energy cooperative with ~30% participation in t community share class (CTH proposal, not a contractual figure).

Community committee
Grassroots organization

→

Energy cooperative
Legal personality

Transition shown

- General assembly — supreme social body
- Board of directors — oversight and strategy
- Operator — technical O&M of the asset
- Energy-sector ally — know-how (role per local agreement)

ajira at scale

ojects in La Guajira (FENOGE and related processes). Pins without official coordinates: *approximate location*.
n: `docs/energy-coop/demo-cth-brand/SOURCES.md`.

Guajira 63.1% (2024)
MPE 2025 (PDF) · PARTIAL — 1ST DEPT. SOURCE

ICEE La Guajira 54.28% (UPME 2024)
Boletín ICEE 2024 · UPME, noticia

with firewood/charcoal/waste: 56.6% of people in energy poverty (IMPE 2025)
MPE 2025 (PDF) · Resumen ejecutivo IMPE 2025

on map

0
Verified kWp (sum)

0
Announced MW (utility-scale)

a Guajira at scale



ajira at scale

ojects in La Guajira (FENOGE and related processes). Pins without official coordinates: *approximate location*.
n: docs/energy-coop/demo-cth-brand/SOURCES.md.

Guajira 63.1% (2024)
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with firewood/charcoal/waste: 56.6% of people in energy poverty (IMPE 2025)
MPE 2025 (PDF) · Resumen ejecutivo IMPE 2025

on map

2,102
Verified kWp (sum)

750
Announced MW (utility-scale)

cludes only Maicao (1,102) + Cabo/Media Luna (1 MW = 1,000 kWp). Nazareth: kWp not published. 750 MW project is not added to kWp.

cts on map



Lab — prioritization funnel

nergy Coop pipeline narrows candidates before PASO. Public policy figures and documented cross-checks from
rgy-coop/; wave counts (100 → 20 → 10) are EXAMPLE DATA illustrating the flow.

verse (reference)

rget of **500 energy communities 2024–2025**, 40% ZNI (FENOGE). In La Guajira, IPSE reports **148 EC** and **1,409 families** under
Dec 2024).

nergy-poverty-kpis-2026-11.md · FENOGE · IPSE

E / IPEM cross-check

departmental IMPE incidence (e.g. La Guajira 63.1%, Chocó 52.8%), rural vs urban ICEE, household IPEM (different index). Does
field visits.

k: energy-poverty-kpis-2026-11.md · IMPE 2025 PDF

urce layer (screening)

ar BASE where coverage exists (e.g. Corozal, Uribia); NASA POWER / Global Solar Atlas where not (Quibdó, Bahía Solano). Roof
ground truth.

i-coverage-2026-11.md · solar-api-integration-plan.md

DATA

100

Candidates after public cross-check + preliminary scoring

↓

20

Prioritized after technical and governance validation

— productive community candidates

s in dialogue for productive cooperative models. Locations and candidate status only.

DATE COMMUNITY

l, Sucre

al seat — reference point with Google Solar BASE coverage in map check (9.31778, -75.29583).
api-coverage-2026-11.md

ve communities in dialogue (territorial name only):

DATE COMMUNITY
, La Guajira

CANDIDATE COMMUNITY
Puerto Asís / Guzmán, Putumayo

CANDIDATE COMMUNITY
La Macarena, Meta

DATE COMMUNITY
Chocó

CANDIDATE COMMUNITY
San José del Guaviare, Guaviare

CANDIDATE COMMUNITY
Miranda, Cauca

DATE COMMUNITY
Arauca

cooperative in 3 clicks

e CEOF flow (cooperative structure + investors + operator). CTH DESIGN PROPOSAL

e a community — Select —

size

25 kWp | 50 kWp | 100 kWp

Create cooperative

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Live demo

https://energiacoop.cleantechhub.net/demo/

← → · Space · CleantechHUB branding

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